

Dhruv

dhruvlohab22@gmail.com | +91 87085 32811 | www.linkedin.com/in/dhruv-lohchab | <https://github.com/Dhruv-Lohchab> | dhruv-portfolio-black.vercel.app

EDUCATION

B.Tech. Computer Science Engineering	July 2023 – July 2027
Manav Rachna University, Faridabad	Avg. SGPA : 8.2 (Till 5th Sem)
Class XII	April 2021 – April 2022
Delhi Public School Refinery Panipat	Percentage : 91 %
Class X	April 2019 – April 2020
Delhi Public School Refinery Panipat	Percentage : 90.8 %

SKILLS

- **Languages:** Python (Primary), SQL
- **Technologies & Libraries :** VS Code, Google Colab, Kaggle, Git/GitHub, NumPy, Pandas, FAISS, Sentence Transformers (BERT)

PROJECT

- **WiSearch**
 - **Project Overview & Problem Solved:** Developed an Intelligent Semantic Search Engine to solve the "vocabulary mismatch" problem in academic research, enabling the retrieval of contextually relevant literature based on user intent and meaning rather than exact keyword matches.
 - **AI & Backend Architecture:** Engineered a high-performance Python/FastAPI microservice utilizing Sentence Transformers (all-MiniLM-L6-v2) to generate dense vector embeddings, successfully mapping complex research abstracts into high-dimensional semantic spaces.
 - **Search Optimization & FAISS:** Implemented FAISS (Facebook AI Similarity Search) for advanced vector indexing and clustering, ensuring highly scalable, sub-second similarity matching across a curated dataset of foundational AI, ML, and Cloud Computing papers.
 - **Full-Stack Development & Fault Tolerance:** Designed a responsive, modern frontend using React, TypeScript, and Tailwind CSS, featuring a fault-tolerant dual-pipeline architecture that gracefully falls back to a custom simulated hybrid search if the primary Python backend is offline.
 - **Algorithmic Filtering & User Experience:** Wrote custom pre-search validation algorithms to filter out-of-domain queries and implemented adaptive relevance scoring, packaged within a highly polished, accessible UI utilizing Framer Motion for fluid transitions.

TRAININGS AND CERTIFICATIONS

- Salesforce AI with Agentforce Champion Program | *Salesforce (Trailhead)*
- Introduction to Artificial Intelligence | *Linkedin Learning*
- Artificial Intelligence Foundations | *Linkedin Learning*

ACHIEVEMENTS & AWARDS

- Presented with LOA for Co-Head of Dirshti Society @ DSW 2024-25
- Awarded for contributions as Co-Head of Media Team @ MRCPS 2024-25

LANGUAGES

English, Hindi